

Certified Algorithms for the Weighted Feedback Vertex Set and Travelling Salesman Problems

Martin Sokolov

September 15, 2024

Contents

1	Introduction	2
2	Preliminaries	3
2.1	Graph Theory	3
3	Methods	4
4	Results	5
5	Conclusion	6

1 Introduction

This section introduces the topic and provides background information.

2 Preliminaries

In this section, we provide the reader with the necessary information in order to follow the paper. We start off with a reminder on Graph Theory. After that, we move to more advanced definitions in the field of Combinatorial Optimization. Finally, we get more on topic by introducing perturbation resilience and certified algorithms from the field of Beyond the Worst-Case Analysis of Algorithms.

2.1 Graph Theory

In this section, the authors refer to Bondy and Murty [1]. From the same book we get a definition for a *graph* G is an ordered pair $(V(G), E(G))$. The set $V(G)$ is called a set of *vertices* and the set $E(G)$, a set of *edges*. Together with the incidence function ψ_G , that associates each edge with an unordered pair of (not necessarily unique) vertices, we define a graph G . We are going to use n as the cardinality of the set V and m for the set of edges E . With each edge e of G , let there be associated a real number $w(e)$, called its *weight*. Then G , together with the weighting function $w : E \rightarrow \mathbb{R}$, is called a *weighted graph*, and denoted (G, w) .

A graph is can be *embedded in the plane*, or is *planar* if it can be drawn on the plane in a way that edges intersect each other only at their ends. Such a drawing is called a *planar embedding* of the graph. A graph G is called *outerplanar* if it has a planar embedding $\tilde{G}$ in which all the vertices lie on the boundary of the outer face. This is also called a *1-outerplanar graph*. Two forbidden minors of that class are the K_4 and $K_{2,3}$ graphs. In the next subsection we familiarize the reader with Baker's approach [2]. Here we define a *k-outerplanar* to be a graph that has a planar embedding in which the vertices belong to at most k concentric layers. Every planar graph is *k-outerplanar* for some k . Given a graph G , we look for *k-outerplanar* embedding of G for which k is minimal.

3 Methods

This section introduces the topic and provides background information.

4 Results

This section introduces the topic and provides background information.

5 Conclusion

This section introduces the topic and provides background information.

References

- [1] J. A. Bondy, U. S. R. Murty *et al.*, *Graph theory with applications*. Macmillan London, 1976, vol. 290.
- [2] B. S. Baker, “Approximation algorithms for np-complete problems on planar graphs,” *J. ACM*, vol. 41, no. 1, p. 153–180, jan 1994. [Online]. Available: <https://doi.org/10.1145/174644.174650>